UNIVERSITY OF TECHNOLOGY SYDNEY
Faculty of Engineering & Information Technology

31338 Network Servers
32520 Systems Administration

Week 1

Introduction and System Startup

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Subject Information

- Please check Subject Outline on Canvas for details!
- Check Subject Outline -> Assessment
  - Assessment 1 - Skills Test (20%): Week 5 in Lab session
  - Assessment 2 – Lab Tests & Quizes (30%): Weeks 3, 4, 6, 7, 8, 10 and 11
  - Assessment 3 – Final Skills test (40%): Week 12
  - Assessment 4 - Learning Journal (10%): Submitted to Canvas **(Weeks 6 & 12)**

Note: For quizzes and skill tests, you can only use Learning Journal as your reference. You can't do online search and/or read other books.

- Class Info:
  - Lectures are online only.
  - Labs are run On-Campus only in CB11.09.405.
- Slides and Labs:
  - **For labs, you are strongly required to go through the pre-recorded videos!**
  - For online lectures, you will catch up the basic concepts. Then you need to connect them to the lab practices.
  - System: Windows Laptop/PC or Lab PC with CentOS 10 and Windows Server 2025

Typical sysadmin tasks

- Installing & configuring servers
- Installing & configuring application software
- Creating & maintaining user accounts
- Backing up & restoring files
- Monitoring & tuning performance
- Configuring a secure system
- Using tools to monitor security
- Upgrades, security patches
- Fault finding / fixing
- Automating repetitive tasks
- Answering questions
- Negotiating & monitoring service agreements
- Meetings
- Initiating or updating policy documentation
- Purchasing new equipment
- Maintaining awareness of security threats & patches

The Boot Process

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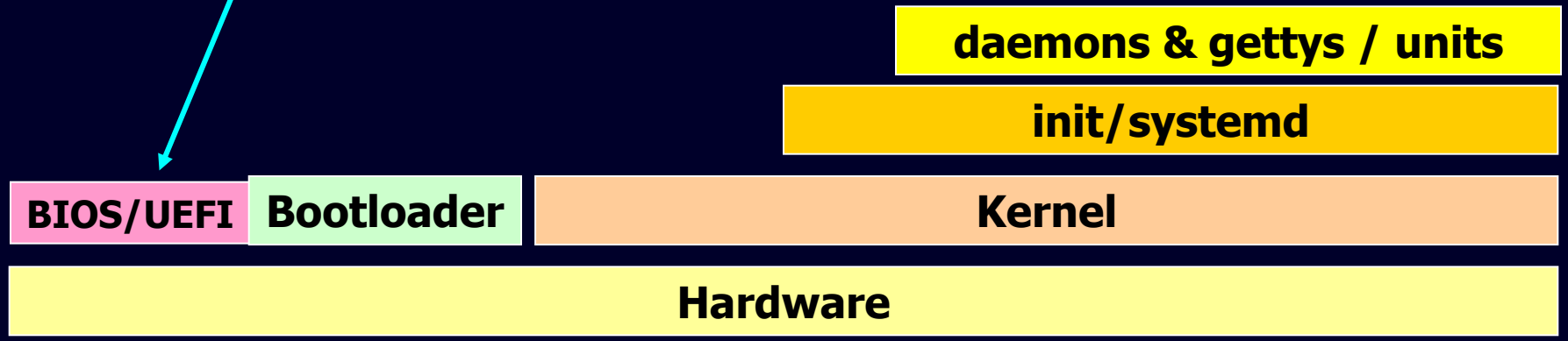
1. The server **firmware** (BIOS/UEFI) starts, performing a quick check of the hardware, called a Power On Self-Test (POST), and then it looks for a boot loader program to run from a bootable device.
2. The **boot loader** runs and determines what kernel to load.
3. The **kernel** program loads into memory; prepares the system, such as mounting the root partition; and then runs the initialization program.
4. The **initialization** process starts the necessary background programs required for the system to operate (such as a graphical desktop manager for desktops, or web and database applications for servers).



BIOS (Basic Input/Output System) and UEFI (Unified Extensible Firmware Interface) are both firmware interfaces that initialize hardware and load the operating system

Firmware: BIOS vs. UEFI

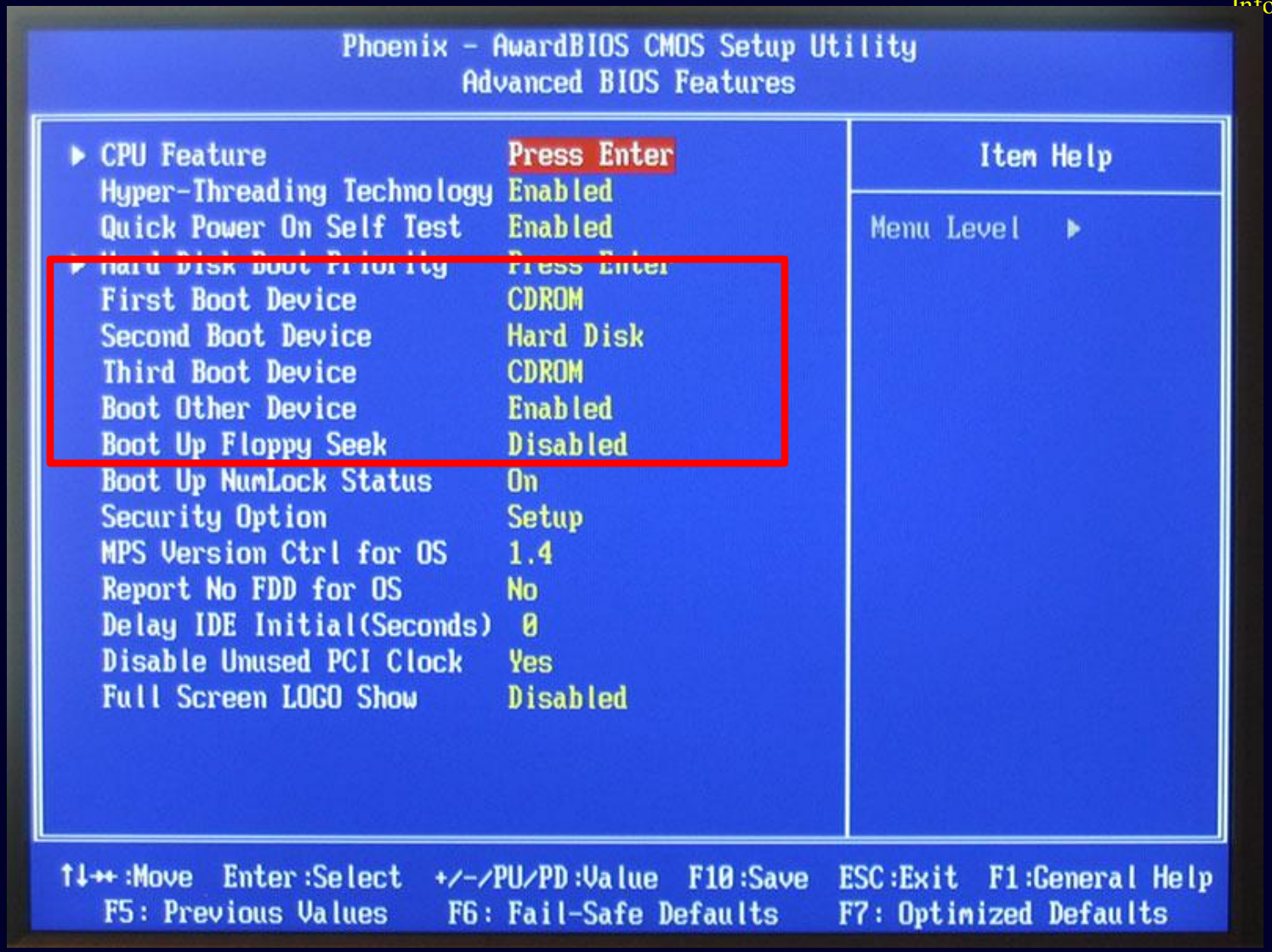
- BIOS = Basic Input/Output System
  –located on ROM chip on motherboard, NVRAM for settings.
- UEFI = the Unified Extensible Firmware Interface, to replace the BIOS
- Most basic function:

  –Execute a power-on self test (POST) and then go to a "known place" known as Master Boot Record (MBR) that is a traditional partition style in computer system and load a bootstrap program into memory and start it running.
- BIOS usually allows configuration of the "known place"

  –e.g. may be a disk sector (hard disk, floppy or CD-ROM)

  –some BIOS/ROM chips can boot from a network address

BIOS


Phoenix - AwardBIOS CMOS Setup Utility
Advanced BIOS Features
CPU Feature Press Enter
Hyper-Threading Technology Enabled
Quick Power On Self Test Enabled
First Boot Device CDROM
Second Boot Device Hard Disk
Third Boot Device CDROM
Boot Other Device Enabled
Boot Up Floppy Seek Disabled
Boot Up NumLock Status On
Security Option Setup
MPS Version Ctrl for OS 1.4
Report No FDD for OS No
Delay IDE Initial(Seconds) 0
Disable Unused PCI Clock Yes
Full Screen LOGO Show Disabled
Item Help
Menu Level
↑↓→←:Move Enter:Select +/-/PU/PD:Value F10:Save ESC:Exit F1:General Help
F5: Previous Values F6: Fail-Safe Defaults F7: Optimized Defaults

Unified Extensible Firmware Interface (UEFI)

- Created by Intel in 1998 and became a standard in 2005
- Specify a special disk partition called EFI System Partition (ESP) to store boot loader program (can be any size)
- ESP can store multiple boot load programs for multiple OS
- ESP normally mounted in `/boot/efi` directory and boot loader files are stored as `.efi`
- Most of time, backwards compatible

Note: Extensible Firmware Interface (EFI)

To check the /sys/firmware directory and see if the efi directory is exist.
If not, using traditional BIOS to boot the computer

Boot loader

- Most basic function:
  - find the kernel and run it, a bridge between system firmware and kernel
- Boot loaders support multiple kernels
  - e.g., dual-boot machines that can boot Windows and Linux
  - e.g., Linux machines when you are installing a new kernel
- Boot loader is in Master Boot Record (MBR) for BIOS adopted, otherwise, EFI System Partition (ESP) for UEFI

- Main boot loaders in Linux on PCs:
  - LILO = Linux Loader (old, only found on ancient installations, discontinued in Dec 2015)
  - GRUB Legacy = Grand Unified Boot Loader (created in 1999)
  - **GRUB2 created in 2005 as a total rewrite of GRUP Legacy**

Boot loader – GRUB2

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- GRUB2 configuration file: **`grub.cfg`**. Saved in **`/boot/grub`** or **`/boot/grub2`** for BIOS, and **`/boot/efi/EFI/distro-name`** for UEFI

- Sample GRUB 2 configuration file

```
[…]
menuentry "CentOS Linux" {
[…]
        set root=(hd1,1)
        linux16 /vmlinuz[…]
        initrd /initramfs[…]
}
menuentry "Windows" {
        set root=(hd0,1)
[…]
```

1st partition in the 2nd hard drive

1st partition in the 1st hard drive

Boot loader – GRUB2

- Rebuild the configuration file rather than install
  - `grub-mkconfig` or `grub2-mkconfig`. Read configuration files in `/etc/grub.d` and assemble the commands into `grub.cfg`
  - Update the configuration file manually through
  - `grub2-mkconfig > /boot/grub2/grub.cfg`

Troubleshooting startup problems

- Boot into single user mode (described later)
- Investigate key configuration files and fix errors
- Reboot and test

– "booting into single user mode" requires interacting with grub at boot time

Single-user mode limits access to your company file to one person at a time. With more than one user license, you can switch to multi-user mode where you can log in to a company file on different computers at the same time.

Boot loader – Windows

- BOOTMGR looks for active partition
- BOOTMGR reads the BCD (boot configuration database) file from the **`\boot`** directory
  - The BCD contains configuration parameters (older windows used **`\boot.ini`**)
- BOOTMGR executes Windows Loader (**winload.exe**)
  - or winresume.exe when resuming hibernated system
  - Winloader loads drivers & transfers control to the windows kernel

https://learn.microsoft.com/en-us/windows-hardware/drivers/bringup/boot-and-uefi

Q&A

- What program does a system's firmware start at boot time?
  A. A boot loader
  B. The BIOS program
  C. The UEFI program
  D. The POST command
  E. The init program

init/systemd for system boots

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- **`init/systemd`** is the first process ever to run in a UNIX system
  - always has Process ID (PID) of 1. (hint: using **`top`** to check the PID)

- **`Sys V init`** processes instructions from **`/etc/inittab`**
  - typically starts system daemons & gettys
    - daemons are background processes that perform system tasks
    - gettys "get ttys" are processes that monitor terminals for activity (logins); One getty process serves one terminal.
  - may also start a graphical login screen (XDM: X Window Display Manager)

- **`systemd`** (2010+, 'd': daemon), the most popular now for SB
  - Reduce initialization time by starting services in a parallel manner
  - **`which init`** #find the init program file location
  - **`readlink -f /usr/sbin/init`** #check the init program for links
    #/usr/lib/systemd/systemd in our lab

In Linux, a daemon is a background process that runs without direct user interaction, typically performing system tasks or providing services

systemd

- Unit Files rather run a lot of daemons!
  - A unit defines a service, a group of services, or an action. Each unit consists of a name, a type, and a configuration file.
  - Units are identified as **`name.type`**.
- **`systemctl`** utility to manage system services
  - **`systemctl`** COMMAND UNIT-NAME
  - COMMAND=`disable, enable, restart, start, status, stop, reload`
  - COMMAND=`is-active, is-enabled, is-failed`
  - UNIT-NAME: `firewalld, ntpd, dhcpd`
- Category 1: Service Unit Files **`.service`**
  - Contains information such as which environment file to use when a service must be started, located in different directories
  - **`systemctl list-unit-files`** #show unit file name and state (e.g., enabled, disabled, static, etc.)
  - **`systemctl cat xxx.service`** #show more information

systemd

- Category 2: Target Unit Files `.target`
  - Group together various services to start at system boot time.

    `systemctl list-units` #check running units

    `systemctl get-default` #graphical.target

    `systemctl cat graphical.target`

    obtain, set and jump between system targets

    COMMAND=`get-default, set-default, isolate (jump)`
    - `poweroff.target` – shuts down and power offs the system
    - `rescue.target` – also for system recovery – requires root password
    - `multi-user.target` – multi-user mode with no graphical login
    - `emergency.target` – used for emergency system recovery
    - `graphical.target` – full graphical mode
    - `reboot.target` – shuts down and boots the system again
  - A target can be a part of another target, e.g., graphical.target includes multi-user.target which depends on basic.target and others

    `systemctl list-unit-files --type=target` #overview all targets

Sys V init

- Sys V `init` processes instructions from **`/etc/inittab`**
- Linux (and Sys V UNIX systems) has conventionally 7 runlevels (more up to 10 in some cases)
  - A runlevel defines the state of the machine after boot.
  - Only one runlevel is executed on startup; run levels are not executed one after another.

Runlevel purposes and systemd-targets

RUNLEVEL	PURPOSE	SYSTEMD-TARGETS
Runlevel 0	Shuts down the system (halt mode)	poweroff.target
Runlevel 1	Single-user mode	rescue.target
Runlevel 2	Multiuser mode without networking (no network file system)	multi-user.target
Runlevel 3	Multiuser mode with networking (normal mode start); standard runlevel for Linux-based systems without GUI	multi-user.target
Runlevel 4	User-definable	multi-user.target
Runlevel 5	Multiuser mode with networking (supports GUI); standard runlevel for many Linux-based systems with GUI and desktop Unix systems	graphical.target
Runlevel 6	Reboots the system to restart it	reboot.target

Managing runlevels and services

Runlevel

```
grep :initdefault: /etc/inittab   #Check default runlevel
 runlevel #Check current runlevel (shows previous and current)
 telinit 4 #Change runlevel
 init 4 #Change runlevel
 telinit q #Re-read /etc/inittab file, change runlevel and use runlevel
  directly to show current runlevel
```

Services

```
service  SCRIPT  COMMAND [OPTIONS]
  COMMAND=restart, start, status, stop, reload
  service cups start # Common UNIX Printing System to act as
  a print server
  service httpd start # to start a web service
```

Log files

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- If something goes wrong during booting, the system logs may help diagnose the problem

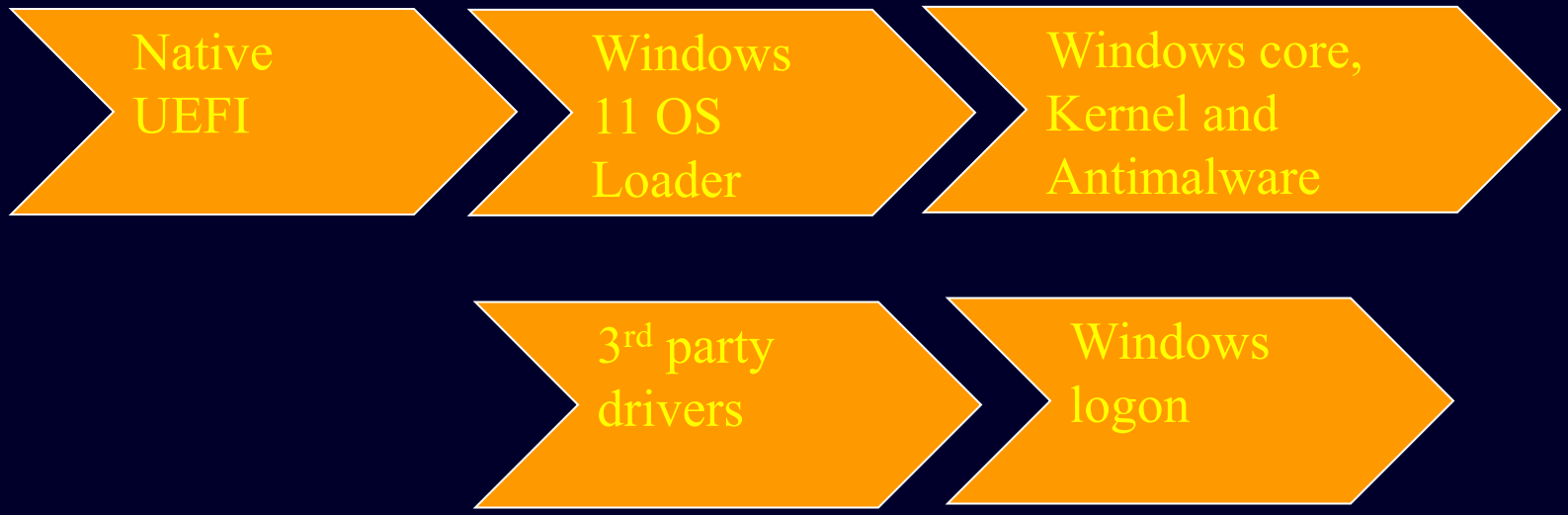
- *Kernel ring buffer* – kernel and module messages
  - `dmesg | less #dmesg: write kernel messages to standard output`
  - (after boot, may be in `/var/log/dmesg` or `/var/log/boot.log`)
  - `journalctl # viewing logs which are collected by system, use -k or -dmesg to retrieve kernel messages`

- Main system log file – **`/var/log/messages`**
  - Contents configured through syslog daemon – more later

- Other log files in **`/var/log`**

Windows startup process

- Windows 11 Startup Architecture



Q&A

- Which of the following is *not* a systemd target unit?
  A. runlevel7.target
  B. emergency.target
  C. graphical.target
  D. multi-user.target
  E. rescue.target

- What memory area does Linux use to store boot messages?
  A. BIOS
  B. GRUB boot loader
  C. MBR
  D. Initrd RAM disk
  E. Kernel ring buffer

Software Installation

- For your own computer, you need to install VMware workstation on PC/laptop. Check the lab setup page in Canvas to install VMs.
- Strongly suggest save VM files* in **SSD portable drive,** hard drive (faster, larger), or USB3.
- A "Virtual Machine" is a pretend computer that runs as a software application on a real (physical) computer
  - It allows you to have a second (virtual) computer inside your real computer
  - "Virtualisation" of servers is touted as a significant saver of money and resources (according to the trade press)

Check the Broadcom webpage for you to install VMware. Set up as a private use.

Please read the section on Canvas – Lab setup – very important! You need to import the two VMs files (.ova) to your SSD drive